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An undergraduate experiment demonstrating the physics of metamaterials with acoustic waves and soda cans

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We describe a novel undergraduate research project that highlights the physics of metamaterials with acoustic waves and soda cans. We confirm the Helmholtz resonance nature of a single can by measuring its amplitude and phase response to a sound wave. Arranging multiple cans in arrays smaller than the wavelength, we then design an antenna that redirects sound into a preferred direction. The antenna can be thought of as a new resonator, composed of artificially engineered meta-atoms, similar to a metamaterial. These experiments are illustrative, tactile, and open ended so as to enable students to explore the physics of matter/wave interaction. © *2016 American Association of Physics Teachers.*
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I. INTRODUCTION

Metamaterials are a growing class of artificial materials that manipulate waves in ways unseen in nature.[1,2] Famous examples include cloaks of invisibility,[3,4] perfect lenses,[5] metasurface holograms,[6] and materials with negative refractive indexes.[7,8] Such materials are made of artificially engineered meta-atoms[9] that mimic the behavior of atoms in solids, but with a much stronger interaction with free-space beams. The meta-atoms provide the flexibility to engineer the effective macroscopic permittivity and permeability of these new materials. Meta-atoms are subwavelength resonators; for example, optical antennas[10,11] absorb and re-emit radiation with a phase and amplitude shift that modifies incident waves via interferences. The collective interaction of these meta-atom arrays create the overall desired effect on freely propagating waves. These metamaterials generally work at microwave and optical frequencies, which involves expensive nano-fabrication techniques and special equipments. Optical/microwave metamaterial experiments for students are therefore difficult to design, despite the principles involved being accessible at an undergraduate level.

In this paper, we present an illustrative undergraduate research project that highlights the physics of metamaterials with acoustic waves and soda cans. At acoustic frequencies, soda cans are Helmholtz resonators,[12–14] and their resonance properties (amplitude and phase) are easily tunable by reducing the volume of air inside the can. Arranged in arrays smaller than the wavelength of sound, soda cans may behave as a collective whole like a metamaterial because the scattering amplitude and phase response can be engineered so as to influence the propagation of sound waves via scattering. To illustrate this principle, we design and demonstrate an acoustic antenna that redirects the emitted sound from a source loudspeaker. We first characterize the resonance amplitude and phase of a single soda can over a frequency range of 280–550 Hz, and retrieve the expected variations of a driven oscillator. We also show that the can's resonance properties are controllable by varying its internal volume. We then arrange two 4×3 arrays of soda cans, set at two different resonance frequencies, so as to amplify the sound from a loudspeaker in one direction and reduce it in the opposite direction. We confirm the antenna's impact on the source by monitoring the radiation pattern of the antenna, and retrieve a cardioid distribution. This pattern is predicted by a computer model of three interfering sources, described later.

The (relatively) large wavelength of sound (on the order of a meter), and the similarity of a soda-can array to a metamaterial, enables students to conduct this experiment on a macroscopic scale with basic audio equipment found in any undergraduate laboratory. This simplicity makes it possible for undergraduate students to investigate the physics of metamaterials, and gives them the opportunity to carry out a research project autonomously with minimal supervision. Such a research project will allow students to gain new insights into the physics of matter/wave interactions, and will complement their classroom learning. Moreover, such an experience will also help students develop the required toolbox necessary to deal with the inevitable hurdles that are encountered when performing research.

II. EXPERIMENT I: A SINGLE CAN

A metamaterial is built of individual "meta-atoms," periodically arranged, that scatter incident radiation. Each meta-atom is smaller than the wavelength of the sound to avoid diffraction effects, and mimic the effect of natural materials on waves. A wave will experience the collective interaction of these meta-atoms interfering with each other rather than individual meta-atoms. In this section, we study the resonance scattering properties of a single meta-atom—in this case a soda can.

A. Theory

Soda cans are Helmholtz resonators.[12–14] The resonances arise from acoustic oscillations of an effective mass of air[15] near the can's opening, sustained by differences in air pressure inside and outside the can, as shown in Fig. 1(a). The pressure differences lead to a restoring force based on the position of the effective mass of air, and the oscillating air results in a sound wave.

The position $X(t)$ of the effective mass of air can be modeled using the equation for a damped harmonic oscillator[16] driven by a pressure difference $\Delta P e^{i\omega t}$

$$\ddot{X} + \gamma_D \dot{X} + \omega_0^2 X = g \Delta P e^{i\omega t}, \tag{1}$$

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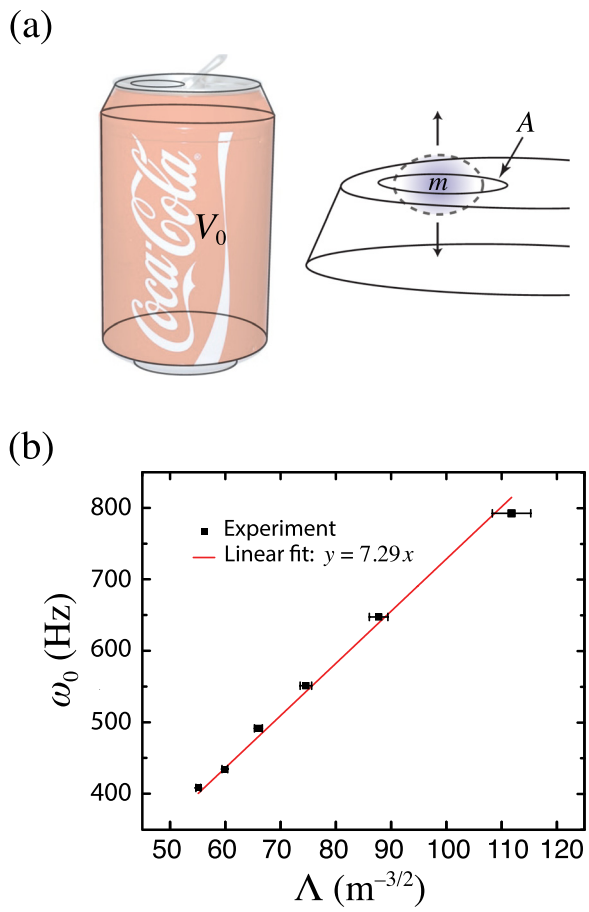


Fig. 1. (a) A soda can is a Helmholtz resonator as the effective mass m of air near its opening oscillates due to the difference in air pressure. (b) Resonance frequency versus $\Lambda = 1/V_0^{-1/2}$ for different internal volumes V_0. The black squares are experimental measurements of the resonance frequency; the line is a linear fit.

where g is a coupling constant between the pressure wave and the air, ω_0 is the resonance frequency, and γ_D is the damping constant. For a Helmholtz resonator, the resonance frequency ω_0 is given by[17]

$$\omega_0^2 = \frac{A^2 \gamma P_0}{m V_0}, \tag{2}$$

where A is the area of the cavity's opening, P_0 is the ambient pressure, m is the effective mass of the air, V_0 is the resonator's internal volume, and γ is the adiabatic constant ($\approx$1.4 for air). Because the oscillator's natural and resonance frequencies are typically close in values, we will consider them to be the same. For a system governed by Eq. (1), the (complex) amplitude is

$$X(\omega) = \frac{g\Delta P}{\omega^2 - \omega_0^2 + i\omega\gamma_D}, \tag{3}$$

and the phase shift ϕ between the motion of the air relative to the incident driving wave is

$$\phi = \tan^{-1}\left(\frac{-\omega\gamma_D}{\omega_0^2 - \omega^2}\right). \tag{4}$$

A soda can partially filled with water has a smaller internal (air) volume, so according to Eq. (2) it will have a higher resonance frequency than an empty can. As described in Sec. II B, we experimentally verify this property by gradually filling the soda can with water and recording its resonance frequency when driven by sound. As shown in Fig. 1(b), the measured resonance frequency is proportional to $\Lambda = 1/V_0^{-1/2}$, as expected. The reader should note that the theoretical model leading to Eq. (2) is only valid under the assumption that the volume of the cavity is much greater than the effective volume of the neck.[15] Deviation of the experimental data from the linear fit are expected for smaller cavity volumes. A similar trend (not shown in this paper) can be observed when the opening of the can is taped over, reducing the can's opening area A.

A variety of related experiments can be performed on a single can. Examples include determining the air pressure or the adiabatic constant, or assessing the volume of the oscillating effective mass in the soda can's neck. In this paper, we make changes to the can's internal volume to control the resonance frequency. Let us now consider how to make these measurements on an individual can.

B. Methodology

A single can is placed an arbitrary distance ($>$ 15 cm) away from a 6-cm-diameter loudspeaker with a microphone held over the can's neck. One approach is to connect the loudspeaker to a frequency generator that produces a sinusoidal waveform of the desired frequency, and then connect the microphone to an oscilloscope to record the waveform, as pictured in Fig. 2. Due to the presence of the can, the recorded waveform is either attenuated or amplified and shifted in time. The frequency is varied from 280 Hz (2,000 rad/s) to 550 Hz (3,450 rad/s) in 20-Hz steps (5–10 Hz steps near resonance) while the amplitude and phase (displacement) of two waveforms—one waveform with the can and one without—are recorded at each step (see Fig. 3). The can's amplitude response is calculated by taking the ratio of the amplitudes of the two waveforms, while the can's phase is calculated from the time shift of the two waveforms.

An extension of this apparatus is to connect both the loudspeaker and the microphone to a computer that is programmed to generate a single short pulse of sound,[13] and then record the temporal response of this pulse. As a

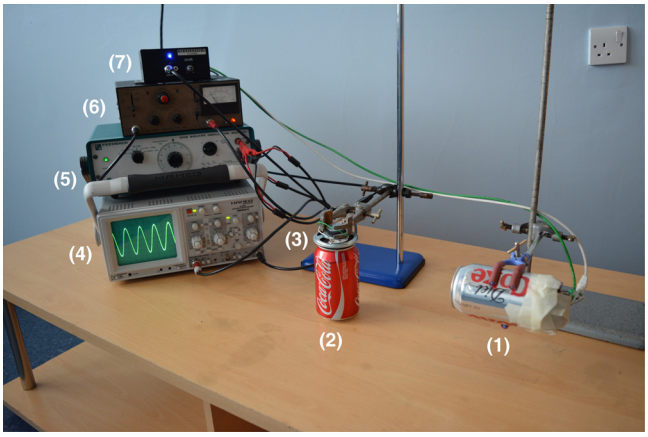


Fig. 2. Experimental apparatus: (1) source (can plus loudspeaker); (2) can under investigation; (3) receiver (loudspeaker used as microphone); (4) oscilloscope; (5) frequency generator; (6) microphone amplifier; (7) source amplifier.

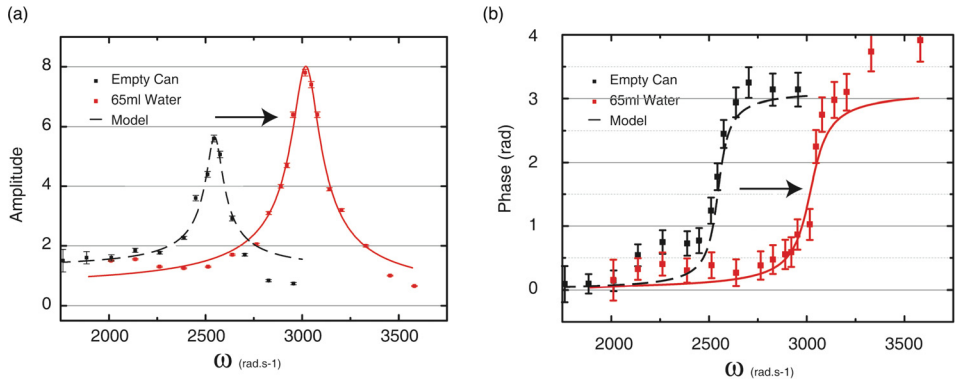


Fig. 3. Can resonance curves: (a) amplitude and (b) phase. The dashed lines is for an empty can ($V_0 = 330$ ml) with resonance peak at ≈2,500 rad/s; solid line is for a can filled with 65 ml of water ($V_0 = 265$ ml) with a resonance peak of ≈3,000 rad/s. The solid curves are the best fits to a single harmonic oscillator.

consequence of the bandwidth theorem—the product of a signal's duration and the range of frequencies within the signal must remain roughly constant—this short pulse will contain a large range of frequencies. A Fast Fourier Transform (FFT) will then extract the can's amplitude response versus frequency. This implementation might be appropriate for a more advanced undergraduate laboratory project.

The reader should note the following points. First, a soda can remains a weak scatterer at acoustic frequencies due to its small size compared to the wavelength. Therefore, the experimental apparatus has been optimized to collect as much signal as possible. Though the presence of the microphone near the can's neck might modify the air flow and alter the inertia of the effective mass of air (and thus the resonance frequency), it ensures a stronger signal to background noise ratio (the noise mainly comes from various room reverberations). Second, our loudspeaker is attached to a soda can solely for amplification around the can's resonance frequency. The presence of the can may change the acoustic spectra of the source, but this does not affect our measurements because, as described below, each measurement is normalized by the source amplitude.

Figure 3 shows the amplitude and phase responses of two soda cans. The first can is empty and has an internal (air) volume $V_0 = 330$ ml, while the second can is filled with 65 ml of water and thus an internal volume $V_0 = 265$ ml. As anticipated, we observe an approximately Lorentzian amplitude response, characteristic of a forced harmonic oscillator, and a phase shift toward higher frequencies. Further measurements up to 830 Hz (5,200 rad/s) show the can's amplitude tends to the background signal, which consists of the incident sound wave and any scattering from the room. Consequently, the can's resonator amplitude must tend to zero and the phase must return to 2π. At higher frequencies, only the incident sound wave is detected by the microphone.

In summary, we have confirmed the Helmholtz resonance nature of a soda can, and highlighted the possibility of tuning the resonance amplitude and phase by varying the can's internal volume. These features provide the flexibility to control the scattering phase of these resonators, which commands how scattered waves interfere with incident waves. To illustrate this concept, we have designed an antenna with multiple cans that reorients the emission of a loudspeaker. These investigations into Helmholtz resonators complement the coursework taken by most students on vibration and wave physics. By applying the driven oscillator model directly to experiment, students can develop an intuition for these fundamental concepts. Moreover, the experimental knowledge gained will provide valuable experience for the remainder of the student's degree program.

III. EXPERIMENT II: MULTIPLE CANS

The emission direction of a loudspeaker can be reshaped using only two Helmholtz resonators that are out of phase. However, due to its small size relative to the wavelength, a single soda can only weakly interacts with sound. To overcome this, we build our meta-atoms with 12 soda cans, grouped in a 4×3 array. This arrangement serves to increase the magnitude of the array's response to a sound wave from that of a single can, while also keeping geometrical dimensions smaller than the wavelength. We then demonstrate how these two arrays, set at different resonance frequencies, can reshape the emission pattern of the loudspeaker.

A. Theory

We assume that our meta-atom behaves as a single entity—a single large Helmholtz resonator—and assume a constant phase across the entire array. Under acoustic excitation, the emitted sound from the array of cans will be described by a spherical wave. Due to the small dimension of the loudspeaker (6 cm in diameter) relative to the wavelength, the source may be approximated by a monopole[18] and, therefore, described by a spherical wave.

Under these circumstances, the pressure variation $\Delta P(\mathbf{r})$ of the array at a position $\mathbf{r}$ far from the emission origin $\mathbf{r}_0$ is expressed by[16]

$$\Delta P(\mathbf{r}) = \mathrm{Re}\left\{ A e^{i\phi} \frac{e^{ik|\mathbf{r}-\mathbf{r}_0|}}{|\mathbf{r}-\mathbf{r}_0|} \right\}, \tag{5}$$

where $k = 2\pi/\lambda$ is the wavenumber, A is the amplitude of the harmonic wave, and ϕ is the phase shift relative to the source driving wave. The antenna consists of three sources: a loudspeaker at the origin ($r = 0$) and two can arrays that scatter the sound from the loudspeaker, placed along the z-axis at locations $z = \pm\delta z$. Figure 4 illustrates this geometry. The total pressure at a position $\mathbf{r}$ far from the loudspeaker is then given by

$$\Delta P(\mathbf{r}) = \mathrm{Re}\left\{ A_0 \frac{e^{ik|\mathbf{r}|}}{|\mathbf{r}|} + A_1 e^{i\phi_1} \frac{e^{ik|\mathbf{r}-\delta\mathbf{z}|}}{|\mathbf{r}-\delta\mathbf{z}|} + A_2 e^{i\phi_2} \frac{e^{ik|\mathbf{r}+\delta\mathbf{z}|}}{|\mathbf{r}+\delta\mathbf{z}|} \right\}, \tag{6}$$

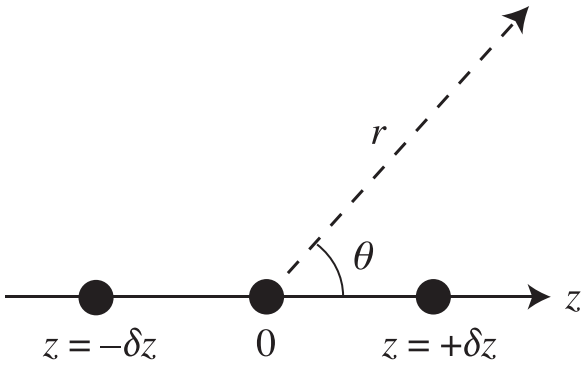


Fig. 4. Illustration of the geometry used to describe three sources.

where $\phi_{1,2}$ represent the soda-can-array's phase relative to the loudspeaker phase, labelled 1 for the positive side ($z>0$) and 2 for the negative ($z<0$).

In the far-field limit ($\delta z \ll r$), we have $|\mathbf{r}\pm\delta\mathbf{z}| = \sqrt{(z\pm\delta z)^2+y^2+x^2} \approx r\pm(z/r)\,\delta z$. Rewriting Eq. (6) in spherical coordinates using $z = r\cos\theta$ then gives

$$\begin{aligned}\Delta P(r,\theta) &= \mathrm{Re}\left\{A_0\frac{e^{ikr}}{r}\left[1+\frac{A_1}{A_0}e^{i\phi_1}\frac{e^{-ik\,\delta z\cos\theta}}{1-(\delta z/r)\cos\theta}\right.\right.\\ &\quad\left.\left.+\frac{A_2}{A_0}e^{i\phi_2}\frac{e^{ik\,\delta z\cos\theta}}{1+(\delta z/r)\cos\theta}\right]\right\},\\ &\equiv \mathrm{Re}\left\{A_0\frac{e^{ikr}}{r}f(\theta)\right\}. \end{aligned} \tag{7}$$

This expression demonstrates that the two arrays modify the loudspeaker's signal in an angular-dependent manner. Assuming $A_1\approx A_2$ and $\delta z \ll r$, the directional function $f(\theta)$ can be approximated as

$$f(\theta) \approx 1+\frac{A_1}{A_0}\left[e^{i(\phi_1-k\,\delta z\cos\theta)}+e^{i(\phi_2+k\,\delta z\cos\theta)}\right]. \tag{8}$$

We can see from this expression that the directional function $f(\theta)$ can be modified by changing the distance $2\delta z$ between the two soda-can arrays or by changing the resonance phases ϕ_1 and ϕ_2 of the arrays. In particular, with the proper choice of ϕ_1, ϕ_2, and δz, the sum of sound waves propagating along one direction ($\theta=0$) can be set greater than the sum of waves propagating in the opposite direction ($\theta=\pi$), creating a directional antenna.

We define the antenna's polarity p as the difference in amplitudes on each side of the antenna

$$p = \Delta P(\delta z,\theta=0) - \Delta P(\delta z,\theta=\pi). \tag{9}$$

Figure 5 shows the polarity for different values of $\phi_{1,2}$ and two different array separations 2 $\delta z = 18$ cm ($\approx\lambda/4$ at 480 Hz) and 2 $\delta z = 36$ cm ($\approx\lambda/2$ at 480 Hz). We observe areas of positive (left and right areas; red online) or negative (top and bottom areas; blue online) polarity indicating a directionality. Because soda cans (as single harmonic oscillators) can only be de-phased by a maximum of π,[16] only the upper-left regions of the plots in Fig. 5 are accessible.

B. Methodology

Our acoustic antenna is built with two 4×3 arrays of cans, as depicted in Fig. 6. The cans are placed on a rotating table, a microphone is placed far from the center of the table, and a loudspeaker (see Fig. 2) is placed above the center of the table. The same equipment as described earlier is used for this experiment. Asymmetric directionality is obtainable by setting the arrays' phases ϕ_1 and ϕ_2 according to the phase maps shown in Fig. 5. However, this approach is quite challenging because it requires precise knowledge of each array's phase. To circumvent this difficulty, we detune the two arrays with a phase shift of π to guarantee that each array has the opposite impact on the source wave—if one array constructively interferes with the loudspeaker's wave, the other will destructively interfere.

A $\pi/2$ phase shift between the two arrays is created by separating the arrays by $\lambda/4\approx18$ cm, as shown in Fig. 6(b). The array-to-array separation is defined by the space between the soda can's neck in the two innermost columns. Another $\pi/2$ phase shift is introduced by setting the arrays at two different resonant frequencies: one set of cans is filled with 65 ml of water, tuning its main resonance to 460 Hz, whereas the second set is filled with 165 ml of water, tuning its main resonance to 520 Hz. The required volumes were found by initially filling two cans with the same amount of water, and then transferring water from one to the other until the two cans were detuned by $\pi/2$. The exact procedure was as

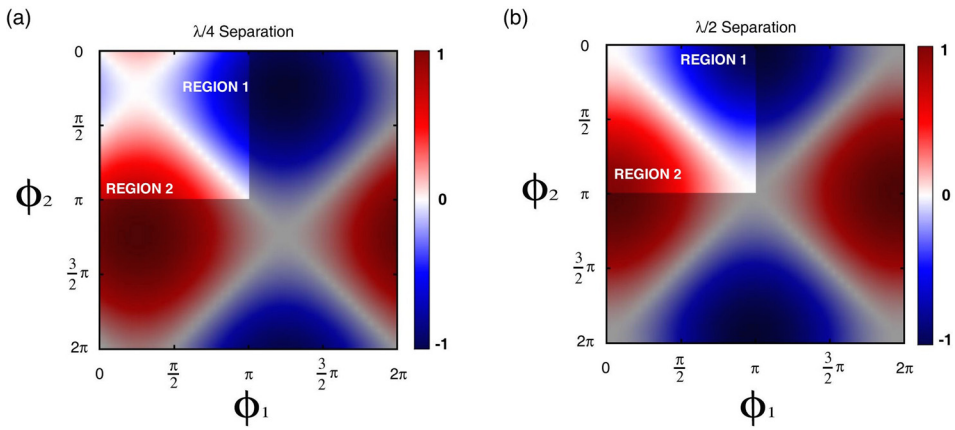


Fig. 5. Antenna's polarity as a function of the array phases ϕ_1 and ϕ_2 in the xz-plane, for two different array separations: (a) 18 cm; (b) 36 cm. The areas labeled "REGION 2" represent positive polarity, while the areas labeled "REGION 1" represent negative polarity; the diagonals (white areas) are neutral. The upper left corners represent the accessible region (a can's maximum phase is π).

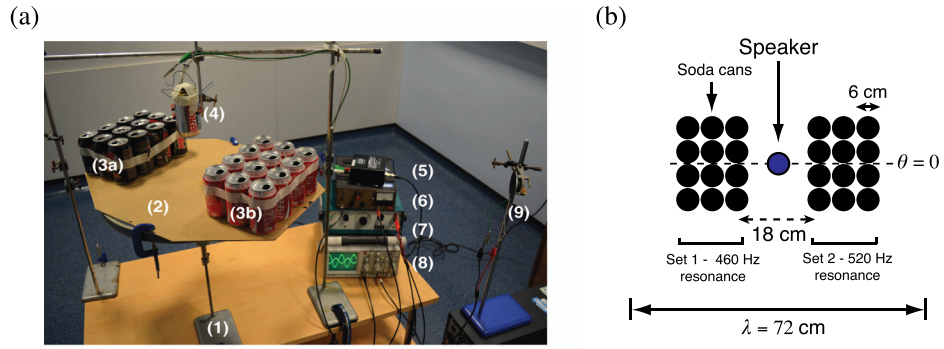


Fig. 6. (a) Antenna apparatus: (1) post; (2) rotating platform; (3 a) and (3 b) cans with 165 ml and 65 ml of water; (4) source (can plus loudspeaker); (5) loudspeaker amplifier; (6) microphone amplifier; (7) signal generator; (8) oscilloscope; (9) receiver (loudspeaker used as a microphone). (b) Schematic of the two arrays, set apart by $\lambda/4$. Notice the antenna is smaller than one wavelength ($\approx$72 cm at 480 Hz).

follows: a loudspeaker was connected to a frequency generator, set to broadcast a sinusoidal signal at 480 Hz. Two soda cans were filled with water so as to resonate at 480 Hz. Two microphones held over the soda cans' openings displayed their resonances onto an oscilloscope. Then, the oscilloscope was zeroed to display the two responses in phase. Water was then gradually transferred from one can to the other can until a phase difference of $\pi/2$ and a similar amplitude were established.

The microphone records the total response of the antenna and source, broadcasting at a frequency of 480 Hz, as a voltage reading on the oscilloscope output. Recordings were made at a distance of 45 cm from the source at 10° increments starting from $\theta = 0$ as defined in Fig. 6(b). The background—defined as the minimum value of the total sound recorded—is subtracted to remove the contribution of the loudspeaker and any reverberations from the room. This procedure isolates the contribution of the two arrays. The normalized polar distributions obtained show a cardioid [see Fig. 7(a)], as expected from Eqs. (7) and (8). We thus see a clear increase of the total sound along the positive z-axis. Uncertainties stem from the angles and the amplitudes measured in this experiment.

The antenna was modeled using the programming language Python at the level of individual soda cans. The

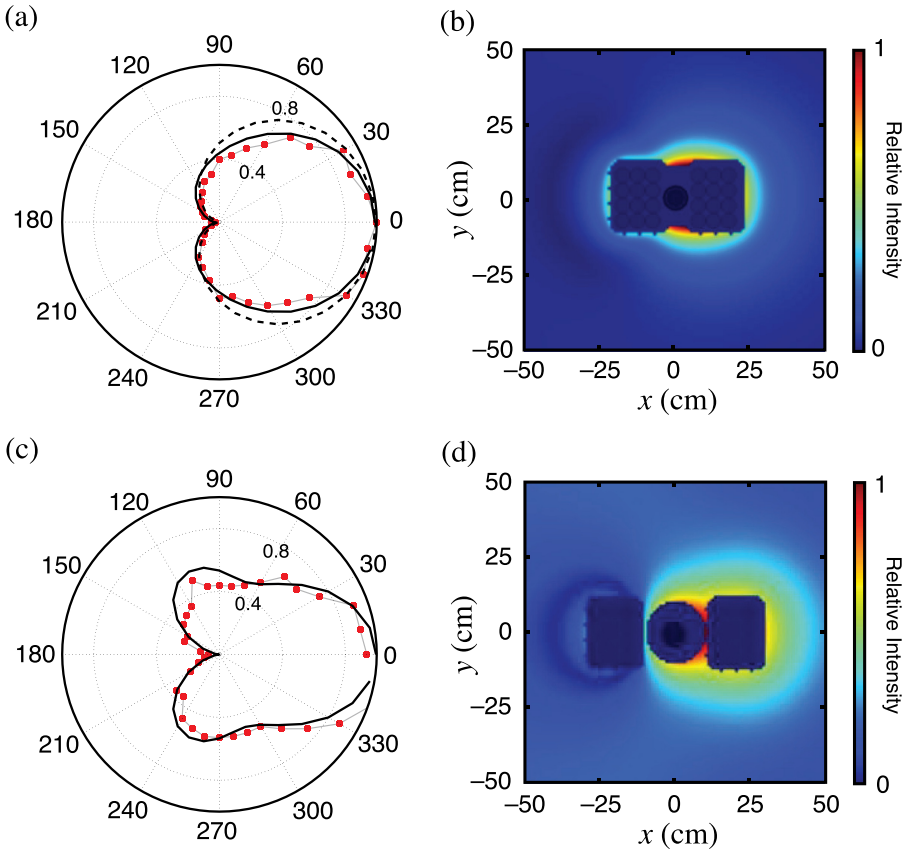


Fig. 7. Background-free normalized polar distributions for an array separation of (a) $\approx$18 cm ($\lambda/4$) and (c) $\approx$36 cm ($\lambda/2$). The dotted lines experimental values, solid curve is the computer model, and dashed curve is the cardioid inferred from our three-sources model without the contribution of the loudspeaker. Simulations of the antenna when each can is modeled as a single harmonic oscillator are shown in (b) and (d). The scale bar represents the relative intensity: maximum is 1, background is 0.

simulation modeled two groups of driven harmonic oscillators (cans), arranged according to the experiment [Fig. 6(b)]. These arrays intercept a 480-Hz spherical driving wave from a source whose amplitude was modeled to decay by the inverse square law, and produce scattering as a result. The resonator's scattering amplitude was calculated using the absolute magnitude of Eq. (3) and the phase from Eq. (4), with a coupling constant $g = 8{,}000\,\mathrm{Pa}^{-1}$ and a damping constant $\gamma_D = 471.23$ m/s, estimated from the width of the resonance in Fig. 3(a). The total antenna's response was simulated over a $2\,\mathrm{m} \times 2\,\mathrm{m}$ space (only $0.5\,\mathrm{m} \times 0.5\,\mathrm{m}$ is shown in Fig. 7) by individually computing the phase and amplitude of each wave (loudspeaker and scattering) in the system at every point in the simulated space, before superposing them to obtain the resultant disturbance.

Interestingly, the phase maps displayed in Fig. 5 also indicate that directionality should be achievable for a separation of $\lambda/2$. Results of this implementation are shown in panels (c) and (d) of Fig. 7. Though the distribution pattern is less like a cardioid, directionality is still maintained, and the polar distribution agrees well with the simulation at the individual-can level.

The small discrepancies observed between the expected distributions and the experiment results could arise from various factors. Though the arrays are smaller than one wavelength, it remains a large scattering object in which phase lags across cans may exist. This affects our choice of spacing of the two arrays. Additionally, due to the large wavelength of the sound, true far-field measurements are difficult to achieve in acoustics as any scattering object (walls, human body, furniture, etc.) can affect the recorded sound. In particular, standing waves appear in small closed rooms, which produce a background signal that modulates the measured sound depending on the location of the microphone. While analyzing the antenna for the first time, we circumvented this problem by placing the antenna in three distinct orientations. Each orientation produced its own set of results while we moved a microphone around the antenna. A control result set was also taken, with no cans present. By comparing these measurements, we identified a definite directional effect far greater than the standing waves detected. From one side of the array to the other, a ten-fold up to thirty-fold increase in amplitude could be found.

Finally, the source and the arrays of soda cans were set at an arbitrary distance varying between 15 cm and 45 cm. Though we are not purely in the far field, this arrangement ensures a stronger signal from the cans' resonances compared to the reverberation of the room. In reality, our measurement includes a mixture of near and far field. Far field is predominant as near field decays as $1/r^3$ in amplitude whereas far field decays as $1/r$. Indeed, the microphone should be placed extremely close to the soda can to see purely near field. Since both near- and far-field components are driven by the same phases from the cans, the interference should be the same and only the precise field distribution should be modified.

IV. CONCLUSION

In summary, soda cans are Helmholtz resonators with resonance properties that can be tuned to create interesting sub-wavelength acoustic devices, for instance, an antenna. The experiment described has illustrated the principle of metamaterials, as it is built out of two sets of sub-wavelength meta-atoms (soda cans) acting in unison to modify sound waves via scattering. The experiment is open ended, as multiple arrangements of the meta-atoms are possible, for instance, designing novel acoustic structures such as a Yagi-Uda antenna,[10] sonic cavities, and even sonic lenses[13,14] with the potential to focus acoustic waves. More advanced follow-up experiments could involve generating short sound pulses using a computer and measuring the temporal response of these arrays.

The physical principles that cause these structures to work are no different to those underpinning optical metamaterials. Yet its implementation only requires undergraduate knowledge and equipment, providing almost complete freedom and autonomy for the student to control and understand any aspect of the experiment. The experiment is divided into a series of investigations of varying complexity, building the student's understanding of the physics as it goes. In particular, this experiment illustrates that the knowledge gained in first-year physics courses are not just pre-requisites for understanding more advanced physics, but also interesting and useful in its own right. It is with this exact undergraduate knowledge that we have investigated the resonance nature of a single can and then built upon that to develop a novel acoustic antenna. For instance, in the first part of the experiment, the relationship between our experimental amplitude and phase data with driver frequency matches the oscillator model taught in undergraduate courses extremely well. This knowledge then provides opportunities for the students to simulate any soda-can acoustic geometry with simple harmonic oscillators via a computer, and explore new research direction by themselves. The second part of the experiment helps teach students how to use information gained in the first part, including the resonance phase of a single can, so as to infer unknown properties of more complex structures, for example, an acoustic antenna. This structured approach provides students experience in how to carry out new research based on previous results.

By leaving the experiment open ended and minimizing supervision, the project described here mimics the true nature of research, encouraging the student to think independently and creatively in order to deal with unprecedented results and unforeseen obstacles. These skills are rarely called upon during traditional undergraduate lab experiments. Yet, such skills are paramount to completing any professional research. With this research project, we hope that the students will satisfy their own curiosity, and find an interest in the novel physics of metamaterials, while building on their understanding of the physics of matter/wave interactions.

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Optics experiments

The picture shows a group of college students doing a set of experiments in optics. The year is about 1950 and the place is the introductory physics laboratory at Kenyon College. In the foreground is a circular optical disk used in a wide variety of experiments in reflection, refraction, image formation and polarization. In the rear a group of students are using a spectrometer, and other students are doing an experiment on image formation. The professor at the left is Elbe H. Johnson, who taught physics at Kenyon from 1914 to 1955. (Notes by Thomas B. Greenslade, Jr., Kenyon College)